Projectile Test-Range Analysis

Collect launch data, model projectile motion, and evaluate how well predictions match the test.

Intermediate	**1-2 class periods**	**Practice & Review**

Engineering Context

A small payload-deployment system must place a sensor inside a target zone. Your team will analyze a safe classroom launcher or simulation.

What You Will Practice

- Measure launch conditions
- Apply horizontal and vertical kinematics
- Create graphs and uncertainty statements
- Compare model predictions with test results

What You Need

- Approved launcher or simulation
- Meter tape
- Angle tool if applicable
- Stopwatch or video timing
- Spreadsheet
- Engineering notebook

Student Instructions

1. Define launch height, angle, and target coordinate system.
2. Record at least five trials at one controlled setting.
3. Calculate predicted flight time and horizontal range using the assigned model.
4. Measure actual landing positions.
5. Calculate average range, range spread, and percent difference from prediction.
6. Graph the landing positions or range results.
7. Identify at least three causes of model-test disagreement.
8. Recommend a setting adjustment for the target zone.

Engineering Requirements

- At least five valid trials
- Consistent launch setting
- All equations and units shown
- Include average and variation, not only one trial
- Recommendation must cite prediction and data

Constraints & Safety

- Use only the approved range and launcher
- No aiming toward people or breakable objects
- Retrieve projectiles only after the range is clear
- Follow all eye-protection requirements

What You Submit

- Test setup diagram
- Calculations
- Trial data
- Graph
- Prediction comparison
- Targeting recommendation

Reflection Questions

1. Which assumption in the model was least realistic?
2. Was the error systematic or random?
3. How would launch-height uncertainty affect the result?
4. What additional measurement would improve the prediction?

Engineering Tip Use a coordinate system and clearly define where horizontal distance begins.	**Common Mistake** Comparing a prediction to the single closest trial hides the actual variation in performance.
Stretch Goal Use your data to create a 90% target interval for expected landing position.	**Career Connection** Flight-test engineers compare analytical trajectories with instrumented tests for release systems, drones, and launch vehicles.